

Maternal & Child Health Analytics Pipeline

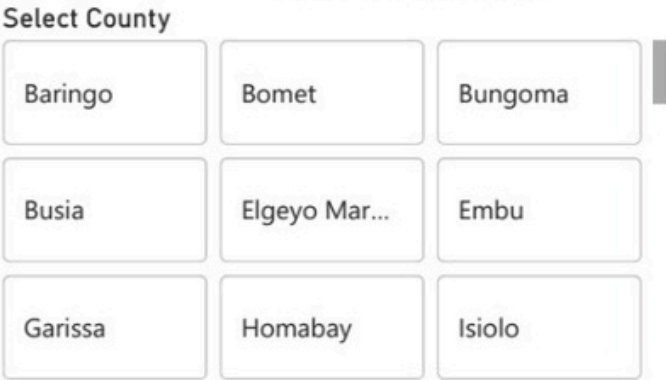
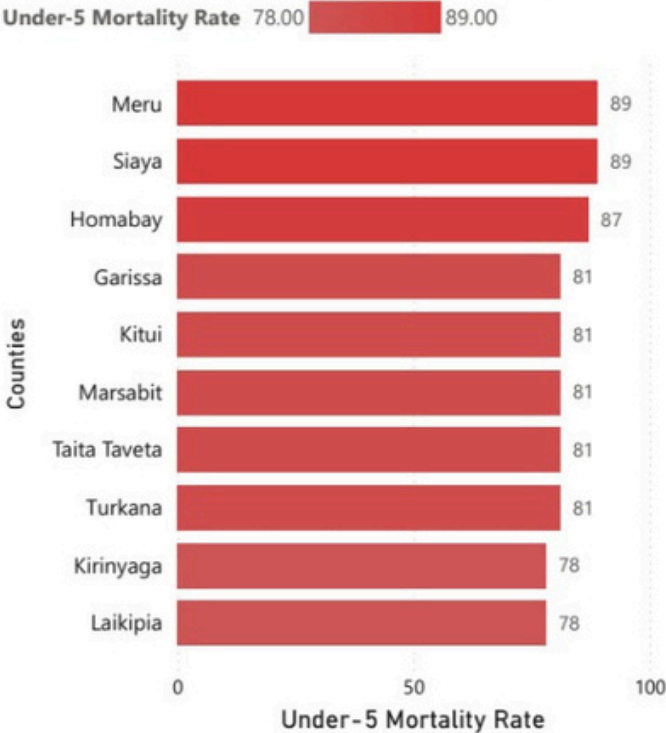
County-Level Maternal Health Monitoring & Decision Support Dashboard

KENYA COUNTY-LEVEL MATERNAL & CHILD HEALTH PERFORMANCE DASHBOARD



Counties	Antenatal 4+ Visits (%)	Postnatal Care (48hrs)	Facility Delivery (%)	Skilled Birth Attendance (%)	Under-5 Mortality Rate	Maternal Mortality Ratio
Baringo	76.00	66.00	86.00	61.00	75.00	652.00
Bomet	78.00	58.00	52.00	67.00	38.00	407.00
Bungoma	70.00	61.00	77.00	86.00	63.00	351.00
Busia	48.00	36.00	92.00	90.00	51.00	253.00
Elgeyo Marakwet	40.00	33.00	77.00	72.00	21.00	503.00
Embu	62.00	68.00	45.00	92.00	28.00	346.00
Garissa	48.00	43.00	70.00	60.00	81.00	386.00
Homabay	49.00	57.00	81.00	91.00	87.00	619.00
Isiolo	87.00	55.00	73.00	50.00	22.00	602.00
Kajiado	52.00	44.00	51.00	66.00	73.00	432.00
Kakamega	68.00	65.00	56.00	55.00	21.00	330.00
Kericho	60.00	72.00	53.00	81.00	20.00	298.00
Kiambu	51.00	80.00	68.00	63.00	71.00	497.00
Kilifi	73.00	30.00	89.00	68.00	47.00	710.00
Kirinyaga	69.00	71.00	88.00	92.00	78.00	350.00
Kisii	48.00	71.00	88.00	76.00	75.00	303.00
Kisumu	82.00	51.00	68.00	94.00	51.00	786.00
Kitui	81.00	56.00	69.00	75.00	81.00	347.00
Kwale	43.00	64.00	60.00	91.00	47.00	397.00
Laikipia	38.00	52.00	83.00	83.00	78.00	236.00
Lamu	52.00	64.00	62.00	54.00	63.00	343.00
Machakos	41.00	38.00	51.00	79.00	56.00	688.00
Makueni	78.00	44.00	53.00	63.00	70.00	750.00
Mandera	43.00	30.00	89.00	58.00	46.00	663.00
Marsabit	60.00	34.00	85.00	62.00	81.00	548.00
Meru	36.00	84.00	59.00	50.00	89.00	545.00
Migori	42.00	31.00	79.00	42.00	74.00	621.00
Mombasa	55.00	56.00	65.00	78.00	25.00	479.00
Muranga	48.00	42.00	52.00	41.00	51.00	614.00
Nairobi	74.00	82.00	66.00	46.00	57.00	426.00
Nakuru	88.00	68.00	85.00	64.00	21.00	359.00
Nandi	88.00	59.00	49.00	51.00	22.00	453.00
Narok	63.00	74.00	72.00	88.00	21.00	208.00
Nyamira	57.00	74.00	84.00	90.00	36.00	453.00
Nyandarua	42.00	44.00	68.00	42.00	63.00	537.00
Nyeri	81.00	55.00	45.00	61.00	43.00	672.00
Samburu	74.00	78.00	52.00	77.00	71.00	545.00
Siaya	65.00	80.00	67.00	83.00	89.00	319.00
Taita Taveta	59.00	76.00	68.00	47.00	81.00	323.00
Tana River	38.00	66.00	91.00	82.00	49.00	400.00
Tharaka Nithi	54.00	43.00	89.00	63.00	46.00	710.00
Trans Nzoia	38.00	81.00	79.00	41.00	31.00	771.00
Turkana	70.00	61.00	55.00	83.00	81.00	462.00
Uasin Gishu	36.00	61.00	79.00	60.00	58.00	201.00
Vihiga	44.00	42.00	78.00	54.00	72.00	603.00
Wajir	84.00	32.00	69.00	78.00	20.00	525.00
West Pokot	84.00	68.00	61.00	69.00	77.00	343.00

Top 10 Counties by Under-5 Mortality (per 1,000)



Insights
>Color scale reflects relative performance across counties
>Higher mortality values indicate increased health system risk



SQL Architecture and Data Lineage

Healthcare KPI View Construction

```
/*
-----
COUNTY KPI VIEW
-----
*/
CREATE OR REPLACE VIEW vw_county_mch_kpis AS
SELECT
  c.county_name,
  f.skilled_birth_attendance_pct,
  f.antenatal_4_visits_pct,
  f.facility_delivery_pct,
  f.post_natal_care_48_hours,

  (100-f.facility_delivery_pct) AS
  facility_delivery_gap_pct,
  (100-f.antenatal_4_visits_pct) AS anc_gap_pct,

  (100-f.post_natal_care_48_hours) AS pnc_gap_pct,

  f.under_5_mortality_rate,
  f.maternal_mortality_ratio,
  f.maternal_risk_level,
  f.under_5_risk_level

FROM fact_mch_county_kpi f
INNER JOIN dim_county c
ON f.county_key=c.county_key;
```

Risk Segmentation Logic

```
RISK SEGMENTATION VIEWS

*/
CREATE OR REPLACE VIEW vw_county_risk_segmentation AS
SELECT
  county_name,
  skilled_birth_attendance_pct,
  antenatal_4_visits_pct,
  facility_delivery_pct,
  post_natal_care_48_hours,
  facility_delivery_gap_pct,
  anc_gap_pct,
  pnc_gap_pct,
  under_5_mortality_rate,
  maternal_mortality_ratio,
  maternal_risk_level,
  under_5_risk_level,

  ROUND((facility_delivery_gap_pct +anc_gap_pct
  +pnc_gap_pct) /3,2)AS service_gap_score,
  CASE
    WHEN maternal_risk_level='High'
    AND under_5_risk_level='High'
    THEN 'Dual Mortality Risk'

    WHEN maternal_risk_level='High'
    THEN 'Maternal Mortality Risk'

    WHEN under_5_risk_level='High'
    THEN 'Under-5 Mortality Risk'

    WHEN (facility_delivery_gap_pct+
    anc_gap_pct+pnc_gap_pct) /3>=35
    THEN 'Service Coverage Gap'
```

Priority Counties Query

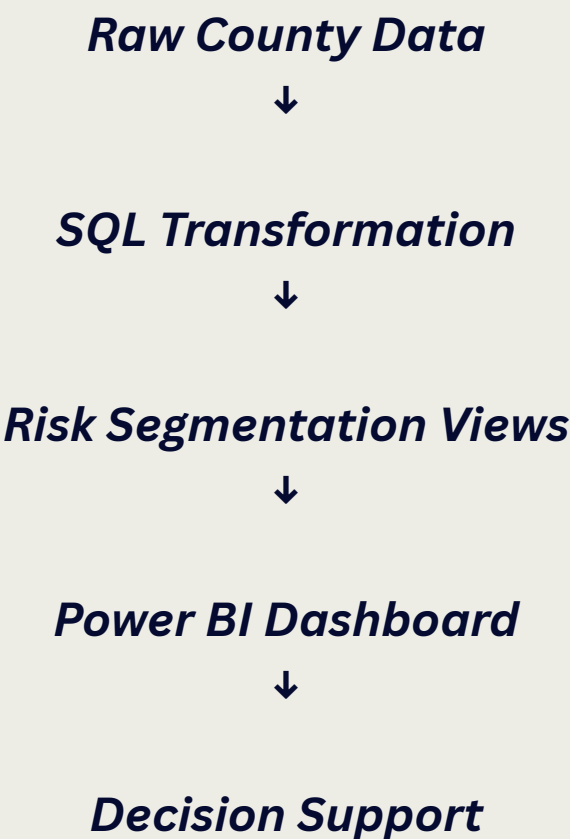
```
/*
-----
PRIORITY COUNTIES FOR INTERVENTION
-----
*/
SELECT
  county_name,
  priority_segment,
  service_gap_score,
  maternal_risk_level,
  under_5_risk_level,
  maternal_mortality_ratio,
  under_5_mortality_rate

FROM vw_county_risk_segmentation

WHERE priority_segment<> 'Monitor'

ORDER BY service_gap_score DESC;
```

Data Lineage Pipeline



County-Level maternal and child health indicators were transformed into reporting ready metrics through SQL views, enabling consistent KPI monitoring across all counties

Maternal Health Service Coverage

Assessment of Care Continuum Performance and Service Delivery Gaps

MATERNAL HEALTH SERVICE COVERAGE ANALYSIS

County-Level Service Coverage Performance

Counties	Skilled Birth Attendance (%)	Facility Delivery (%)	Antenatal 4+ Visits	Postnatal Care (48hrs)
Baringo	61.00	86.00	76.00	66.00
Bomet	67.00	52.00	78.00	58.00
Bungoma	86.00	77.00	70.00	61.00
Busia	90.00	92.00	48.00	36.00
Elgeyo Marakwet	72.00	77.00	40.00	33.00
Embu	92.00	45.00	62.00	68.00
Garissa	60.00	70.00	48.00	43.00
Homabay	91.00	81.00	49.00	57.00
Isiolo	50.00	73.00	87.00	55.00
Kajiado	66.00	51.00	52.00	44.00
Kakamega	55.00	56.00	68.00	65.00
Kericho	81.00	53.00	60.00	72.00
Kiambu	63.00	68.00	51.00	80.00
Kilifi	68.00	89.00	73.00	30.00
Kirinyaga	92.00	88.00	69.00	71.00
Kisii	76.00	88.00	48.00	71.00
Kisumu	94.00	68.00	82.00	51.00
Kitui	75.00	69.00	81.00	56.00
Kwale	91.00	60.00	43.00	64.00
Laikipia	83.00	83.00	38.00	52.00
Lamu	54.00	62.00	52.00	64.00
Machakos	79.00	51.00	41.00	38.00
Makueni	63.00	53.00	78.00	44.00
Mandera	58.00	89.00	43.00	30.00
Marsabit	62.00	85.00	60.00	34.00
Meru	50.00	59.00	36.00	84.00
Migori	42.00	79.00	42.00	31.00
Mombasa	78.00	65.00	55.00	56.00
Muranga	41.00	52.00	48.00	42.00
Nairobi	46.00	66.00	74.00	82.00
Nakuru	64.00	85.00	88.00	68.00
Nandi	51.00	49.00	88.00	59.00
Narok	88.00	72.00	63.00	74.00
Nyamira	90.00	84.00	57.00	74.00
Nyandarua	42.00	68.00	42.00	44.00
Nyeri	61.00	45.00	81.00	55.00
Samburu	77.00	52.00	74.00	78.00
Siaya	83.00	67.00	65.00	80.00
Taita Taveta	47.00	68.00	59.00	76.00
Tana River	82.00	91.00	38.00	66.00
Tharaka Nithi	63.00	89.00	54.00	43.00
Trans Nzoia	41.00	79.00	38.00	81.00
Turkana	83.00	55.00	70.00	61.00
Uasin Gishu	60.00	79.00	36.00	61.00
Vihiga	54.00	78.00	44.00	42.00
Wajir	78.00	69.00	84.00	32.00
West Pokot	69.00	61.00	84.00	68.00

Average Antenatal 4+ visits

59.94

Average Facility Delivery

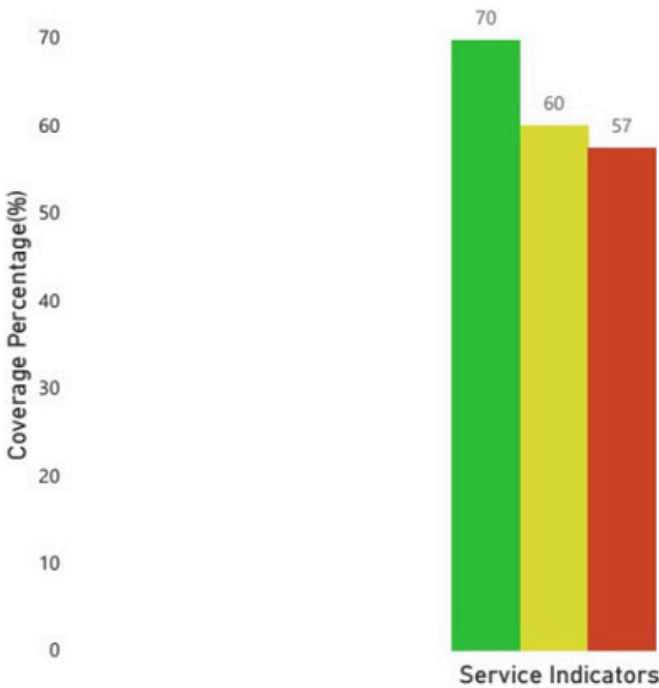
69.74

Average Postnatal Care (48hrs)

57.45

Average Service Coverage Comparison

● Facility Delivery (%) ● Antenatal 4+ Visits ● Postnatal Care (48hrs)



Counties

Baringo

Bomet

Bungoma

Busia

Elgeyo Marakwet

Embu

Garissa

Homabay

Isiolo

Key Insights

- > Postnatal care shows the lowest coverage across most counties
- > Facility delivery rates are relatively higher ,indicating stronger institutional access
- > Antenatal care uptake is moderate but inconsistent
- > Counties with weak postnatal care may face increased maternal and child health risks

The dashboard helps identify service delivery gaps across the maternal care continuum, supporting targeted interventions and resource allocation

Executive Summary & Strategic Insights

Key National Maternal and Child Health Findings and Decision-Support Recommendations

MATERNAL & CHILD HEALTH(MCH) EXECUTIVE SUMMARY

What This Dashboard Helps You Do

- ✓ Stratify Maternal & Child Health Risk Across Counties.
- ✓ Monitor Under-5 Mortality and Service Coverage Trends.
- ✓ Identify Skilled Birth Attendance and Postnatal Care Gaps.
- ✓ Support County-Level Healthcare Decision Making .
- ✓ Transform Raw Clinical Data into Actionable Health Insights.

Average Facility-delivery (%)

69.74

Average Under-5 Mortality Rate (per 1,000)%

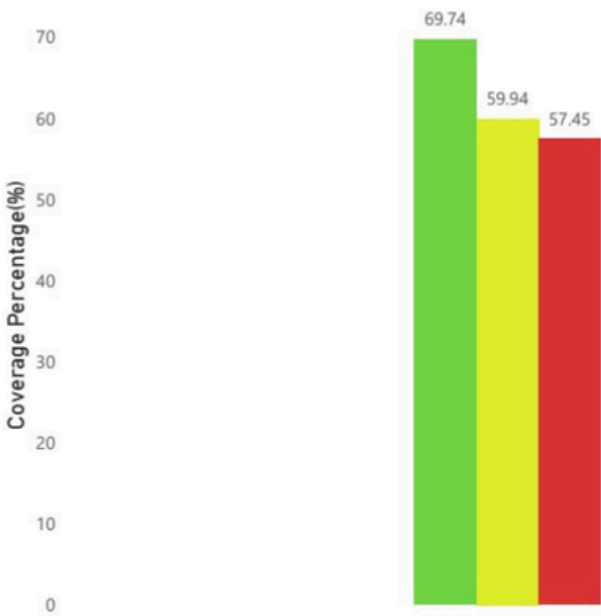
55.34

Average Postnatal Care (48hrs)%

57.45

Average Service Coverage Comparison

● Facility Delivery ● Antenatal Coverage ● Postnatal Care (%)



Executive Insight

Facility delivery coverage remains relatively strong across counties, while postnatal care continues to show measurable service delivery gaps requiring targeted maternal health interventions.

Key Insights

- Counties with higher Skilled Birth Attendance demonstrate lower Under-5 Mortality rates.
- Significant county-level variations indicate differences in healthcare access and service utilization
- Postnatal Care coverage remains the weakest component of the maternal care continuum.
- Several counties exhibit elevated mortality indicators despite moderate service coverage.
- County-level benchmarking enables rapid identification of priority intervention areas.



Healthcare Analytics Pipeline:

MySQL Relational Database → SQL Views → Power BI Executive Dashboard



Power BI

Under-5 Mortality Risk Analysis

Identification of High-Risk Counties and Mortality Drivers

UNDER-5 MORTALITY AND PERFORMANCE ANALYSIS

IDENTIFICATION OF HIGH RISK COUNTIES AND MORTALITY DRIVERS

Average Mortality Rate

55.34

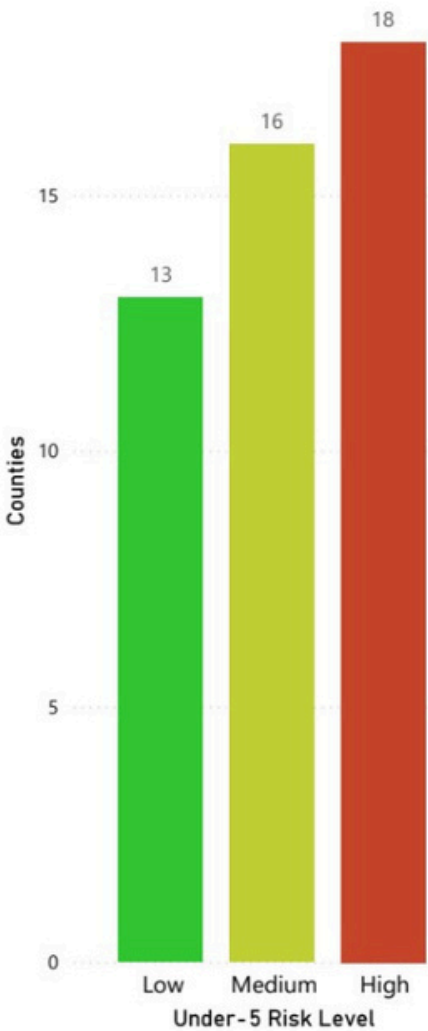
Maximum Mortality Rate

89.00

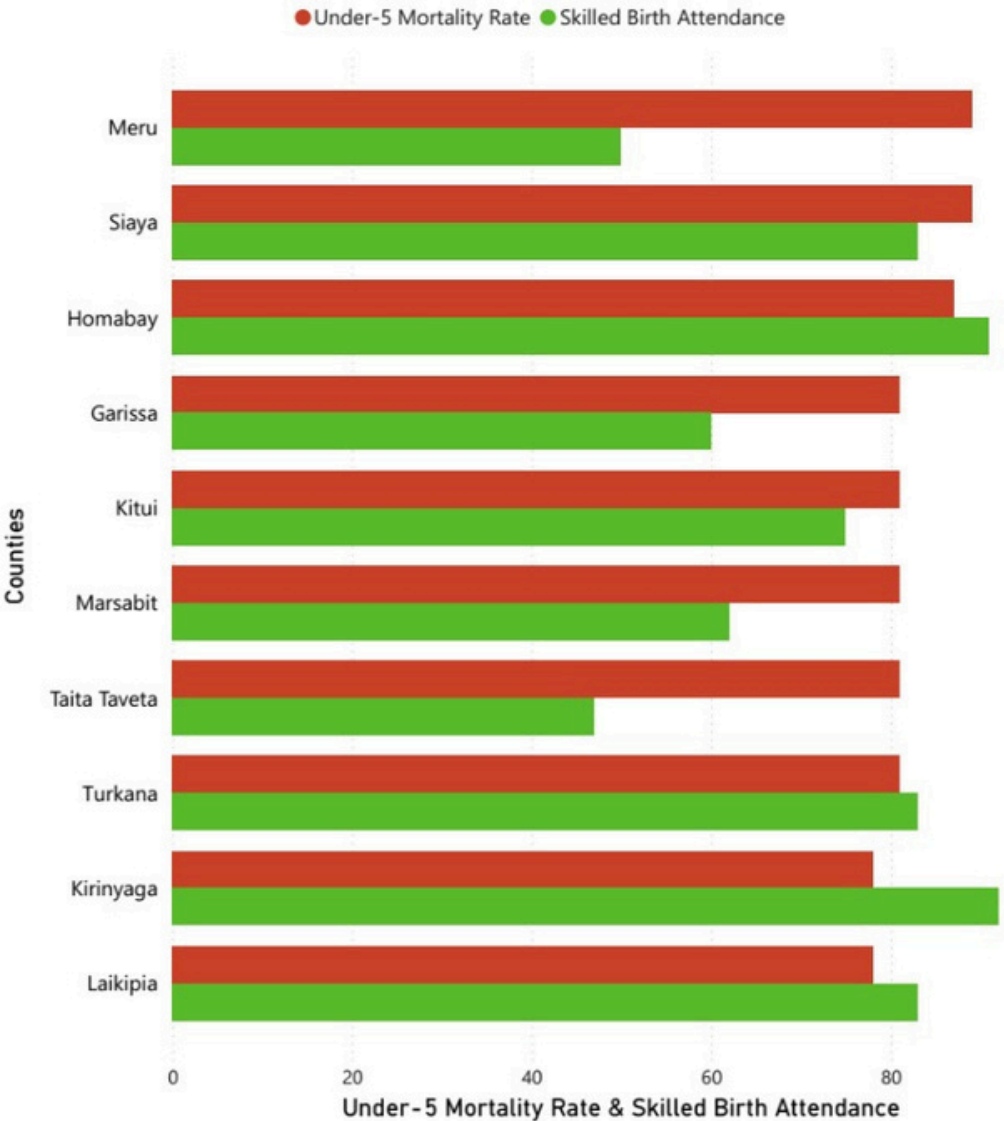
Minimum Mortality Rate

20.00

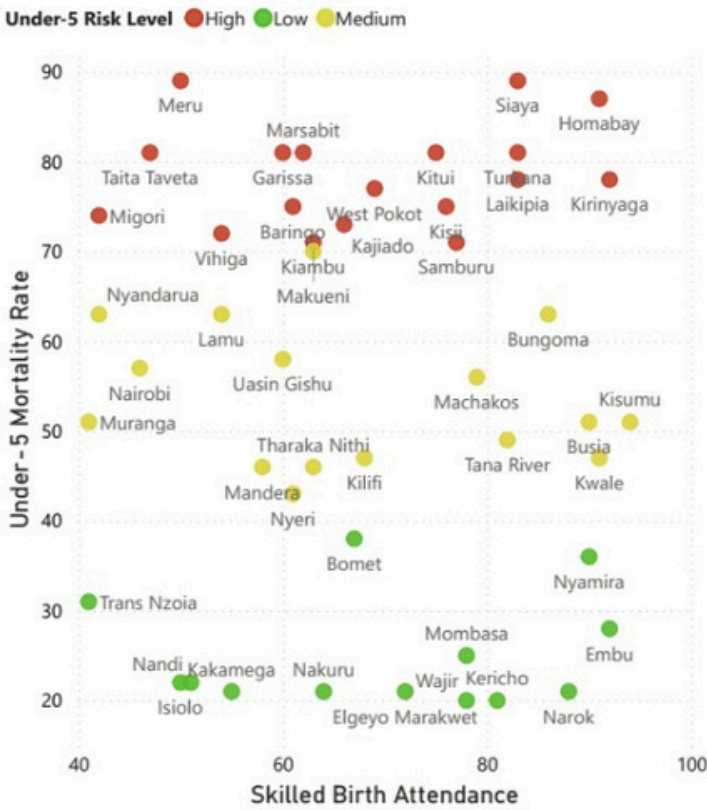
Distribution of Counties by Risk Level



Mortality Rate vs Skilled Birth Attendance (Top 10 High-Risk Counties.)



Mortality vs Skilled Birth Attendance (Risk Segmentation)



Key Insights

- > High mortality counties consistently show lower skilled birth attendance
- > Risk segmentation indicates a concentration of counties in medium to high-risk groups
- > Weak service coverage is strongly associated with increased mortality outcomes

Risk Tiers : High Risk ● Medium Risk ● Low Risk ●

